Amit Kuhrekar also shares his earlier experience of working on a problem to identify challenges for the sachet manufacturing process. In this work process, every machine is a high-speed machine. The sachets that we see in strips are initially formed in a mat, which is later cut. While this happens, the operator needs to stop the machine and check whether the sachet is leaking because of bad thermal sealing. There should not be any obstruction in between the seal for it to stay intact. Even if a single sachet is found to be leaking, the whole sachet batch is kept on hold and then everything needs to be tested manually. To identify if there is a sealing issue and predict and stop the machine, thermal cameras were installed. With the help of ML, thousands of images were generated and specific images were identified.

Further opportunities that AI on edge can provide

In the current Covid-19 scenario, startups can focus on providing augmented reality glasses having low latency and fast video processing that are equipped with thermal cameras. So, even if people are moving, their temperature can be scanned, identified, and an alert can be issued.

When people start returning to offices, there can be another application wherein face recognition technology can scan the identity of the incoming people, thereby removing the need to swipe a card.

So, is it worth putting AI on edge? "Yes!" says Dharmendra Kumar. "AI is capable of making a decision depending on the type of scenario and application. For example, for connected cars, AI can take a decision, but it should also come from the driver. For industrial, heavy-vehicle, railways, and ATC applications, AI can be implemented. On the manufacturing side, manufacturers can implement the intelligence in infra on AI-edge."

Agreeing to that Rishi Gargi says, "It is not necessary to always allow devices to make decisions. Sometimes, it could be a hybrid model to suggest to the user the type of action to take. The device can decide if it has well-defined confidence or prompt the user to take a certain decision. It's a reinforcement kind of situation."

Venkateswar Melachervu points out that everything has a tradeoff. Nonetheless, in terms of being successful on the edge, it is critical to be driven by the utility that users can derive. For that, you could potentially look at the AI-driven CPUs and microcontrollers. Mobile devices these days are pretty much ready in terms of computing resources and storage. TensorFlow Lite provides stacks readily available that can run across platforms like Android and iOS. What you need to look at is whether to optimise on a device itself or do it in an aggregated way in a gateway device within the edge, or do a combination of these two and the cloud at the backend.

When designing the solution, always consider how to move from one technology to another technology, and from one architecture to another architecture, so that you have enough flexibility and robustness.

Gong the AI on edge way?

Overall, both designers and industry leaders can leverage the huge potential that AI on edge offers. So, while the technology brings faster computing speed and intelligent data processing required for solving the modern-day demands, it is also essential to keep in mind the cost of developing it and the enhanced data security of user data.

"Fall in love with the problem and not the solution. If you fall in love with the problem, then you can always find a solution—whether it is on edge or cloud," advises Amit Kurekar.

"There is a strong case for AI on edge. There are MPUs and bionic chips that are making this possible. At the very base level, the hardware is important for doing AI at the chip level. And since they are cheap, computers are also becoming cheap. At the same time, you have to be careful whether you want to put everything at the cloud level or aggregate it at the edge level. On top of it sits the OS that has been built to run AI. Along with this sits the framework layer, which allows you to run optimally on the edge. The programmer is offered a choice of the framework and to build an AI application. Further, on top, you can use Python or C++," says Nikhil Bhaskaran. END

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The article was originally published in the January 2021 issue of Electronics For You